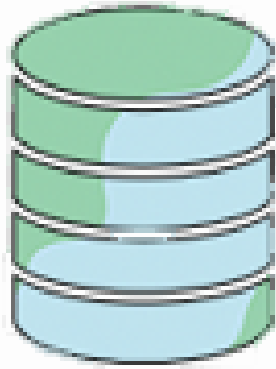


Geodatabase

Design & Management



Information Technology Institute | Geoinformatics Department

Kareem Ashraf



Outline | Introduction to Geodatabase



01

**Geodatabase
Definition**

02

**File-based vs RDBMS
database**

03

**File Geodatabase
in ArcGIS Pro**

04

**File Geodatabase
Size & limits**

05

06

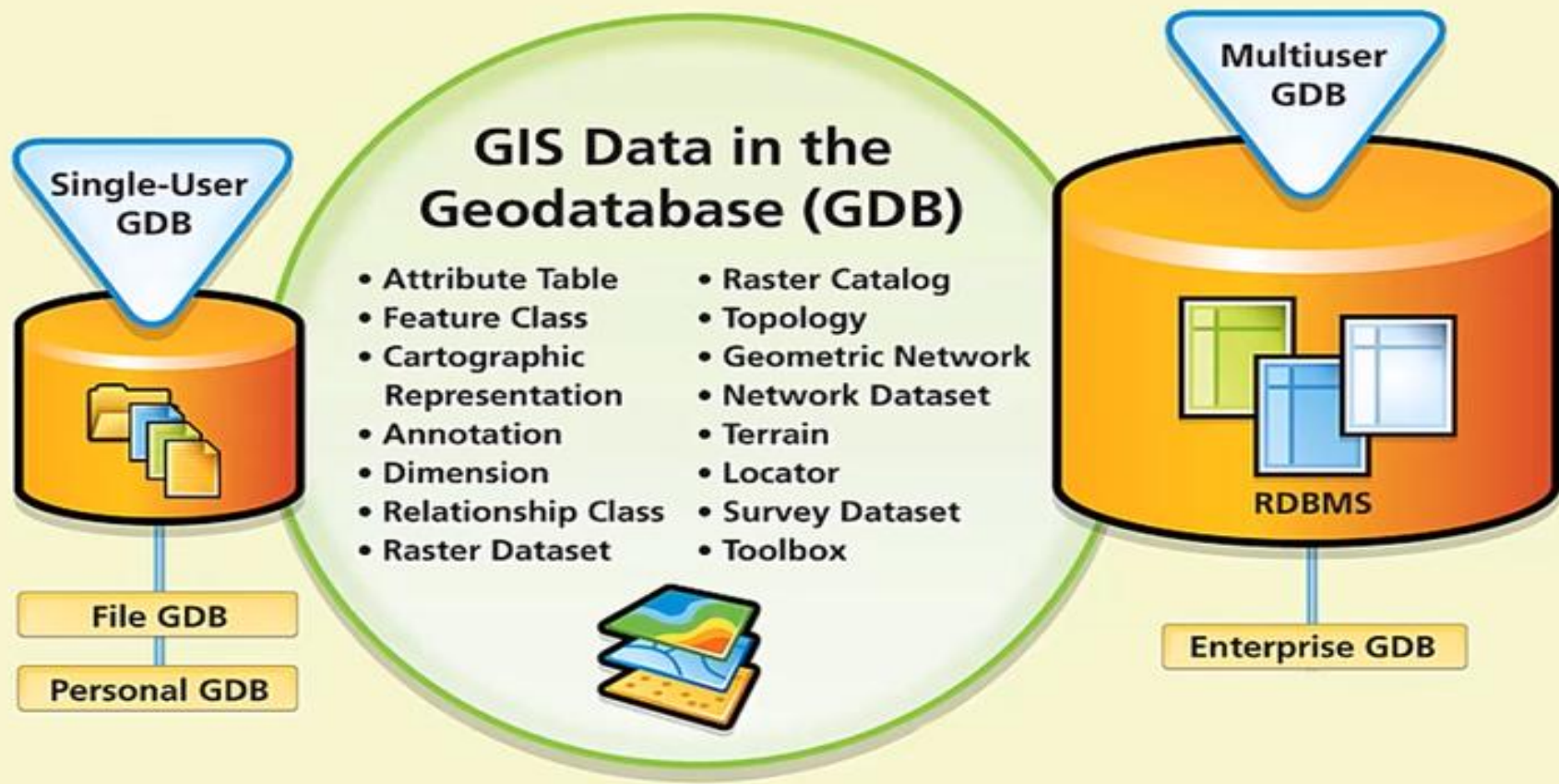
Geodatabase Definition

Geodatabase Definition

- A geodatabase is an organized data storage system used to store, manage, and maintain geographic and attribute data. It provides a structured container for various types of geographic datasets, including feature classes (points, lines, and polygons), raster data, tables, relationships, and topologies
- Geodatabases are crucial to contemporary GIS workflows because they facilitate improved data sharing, versioning, querying, and analysis.
Different kinds of geodatabases exist, including

- **Personal Geodatabase** : built on Microsoft Access Commonly used in ArcGIS Desktop For simpler tasks and smaller datasets (.mdb)
- **File-based geodatabases** : File-based database , stored on disk as files, are useful for editing by a single person and for larger datasets.
- **Enterprise database** (RDBMS) : Commonly used in large organizations and enterprise GIS environments. Supports multi-user editing, large datasets, versioning, high security, and advanced data management for complex and mission-critical applications





File-based *vs* RDBMS database

File-based VS RDBMS database

RDBMS (Enterprise)	File-based (.gpkg - .gdb)	Aspect
Database server	Local folder on disk	Storage
PostgreSQL, SQL Server, Oracle	.gpkg - .gdb	File Type
Multiple concurrent users	Single editor	Users
Supported	Not supported	Concurrent Editing
Excellent for large datasets	Good for small-medium data	Performance
Advanced (users & roles)	Basic (file-level)	Security
Enterprise & government systems	Learning, Individual projects, data exchange	Use Cases

File Geodatabase in ArcGIS Pro

File Geodatabase in ArcGIS Pro

In ArcGIS Pro, a File Geodatabase is a file-based geodatabase format that stores GIS data (such as feature classes, rasters, and tables) within a single folder on disk with a .gdb extension.

It serves as the default data storage mechanism in ArcGIS Pro, providing a structured and efficient environment for managing spatial and non-spatial data.

Although it appears as a single folder to users, it internally contains multiple binary files optimized for performance and data integrity

File Geodatabase Size & limits

File Geodatabase Size & limits

File geodatabase size and limits are as follows :

- File geodatabase size **No limit**
- Table or feature class size **1 TB (default)** - 4 GB or 256 TB with keyword
- Number of feature classes and tables **2,147,483,647**
- Number of **fields** in a feature class or table **65,534**
- Number of **rows** in a feature class or table **2,147,483,647**
- **Feature dataset** or **Feature class** or **table** name length 160 characters
- **Field** name length 128 characters
- Geodatabase name length The number of characters the operating system allows in a folder name

Geodatabase Fundamentals Quiz

Scan the QR Code to start the assessment



→ [Link](#)

Outline | File Geodatabase Structure & Components



01

**Raster dataset
& Mosaic dataset**

02

**Feature Class
Feature dataset
Shapefile**

03

**Import / Export Features
Data Conversion**

04

**Feature Layer
Attribute Table**

05

**Spatial & none
Spatial tables**

06

**Join & Relate
Relationship Classes**

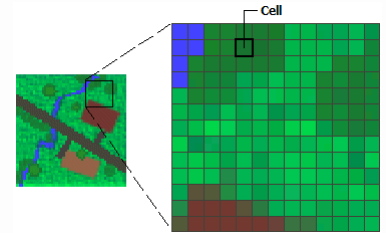
Raster dataset & Mosaic dataset

- Raster dataset 
- Mosaic dataset 

Raster data in File Geodatabase

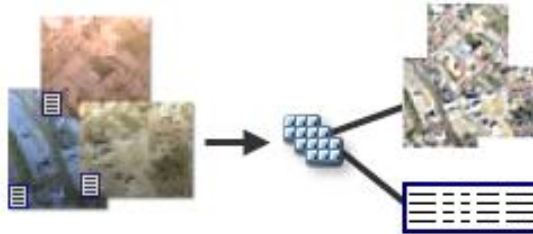
Raster dataset

- A raster dataset is a grid-based data storage model used in the Esri ArcGIS system to store and manage imagery, elevation models, and other continuous data.
- It consists of one or more bands, where each band is an array of pixels, and each pixel stores a numeric value
- ArcGIS Pro supports more than 70 raster formats, including TIFF, PNG , JPG



Mosaic dataset

- A mosaic dataset is a collection of raster datasets stored as a collection of images and viewed or accessed as a single mosaicked image



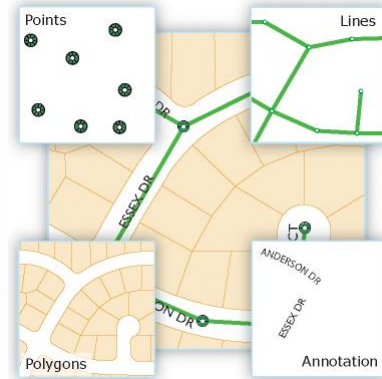
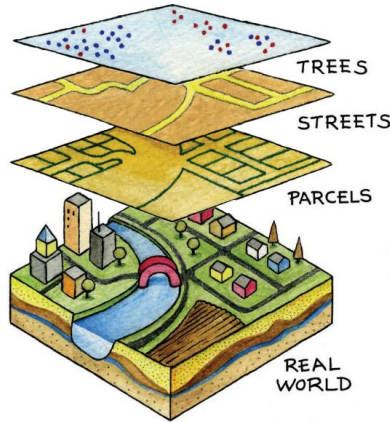


Feature Class

Feature dataset

Feature Classes Definition

- Feature class is a fundamental data structure used to store geographic features that share the same geometry type and spatial reference.
- It stores both geometry (**points**, **lines**, or **polygons**) and attribute data together in a **geodatabase**
- Feature classes are created and stored in a **geodatabase** either as **stand-alone** objects or organized into a feature dataset



Feature Classes Creation

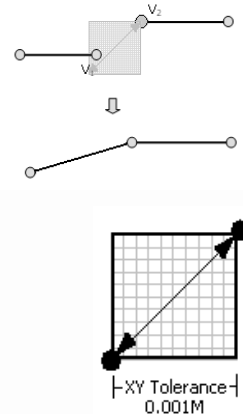
There are four primary ways to create a feature class :

- Using the Create Feature Class wizard
- Using the Create Feature Class geoprocessing tool
- Exporting a map layer to create a feature class
- Converting an external data source, into a feature class

Create Feature Class wizard

The Create Feature Class wizard consists of six pages , Each page defines a specific aspect of the feature class

1. **Define Page** : In this page, you define the feature class **name** and **alias**, select the **geometry type**, and set the **geometric properties** that control whether the feature class supports **Z** and **M** values
2. **Fields Page** : The Fields page is used to **design the attribute structure** of the feature class. In this page, you define the fields that will describe the properties of each feature, including field names, data types, and basic field rules, Fields can be created manually or imported from an existing table or feature class
3. **Spatial Reference Page** : In this page, you can choose the required **coordinate system** to the feature class
4. **Tolerance Page** : It defines the minimum distance at which two points are considered distinct helping manage snapping behavior and reduce geometric errors during editing
5. **Resolution Page** : defines the precision used to store coordinate values. It determines how many decimal places are retained when storing feature coordinates, directly affecting spatial detail and accuracy
6. **Storage Configuration Page** : defines how the feature class is physically stored in the geodatabase. These settings are mainly relevant for enterprise geodatabases and control performance and storage behavior



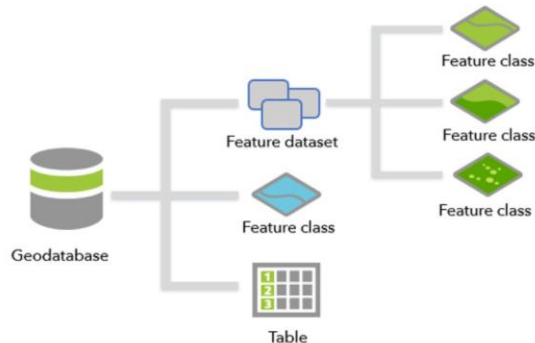
Feature dataset

container within a geodatabase that holds related feature classes that share the same coordinate system, It enables advanced data management by supporting complex datasets such as Topologies, network datasets to ensure data integrity and analysis

Key Characteristics :

- **Organization:** Acts like a folder to group thematic data (e.g., all utility features).
- **Shared Spatial Reference:** All feature classes inside must use the same coordinate system.
- **Enables Advanced Rules:** Essential for implementing advanced datasets like **Topology, Network Datasets**

In RDBMS Only The owner of the Feature data set can add new Feature class to it





Shapefile

Shapefile

A shapefile is a vector data storage format used to store the location, shape, and attributes of geographic features that share the same geometry type and spatial reference. It is not a database, but a collection of multiple related files stored together on disk, where each file plays a specific role in representing the dataset

1. **.shp** : Main file that stores the geometry of features (points, lines, or polygons).
2. **.shx** : Index file that links the geometry to the attribute records.
3. **.dbf** : Attribute table that stores non-spatial data for each feature.
4. **.prj** : Projection file that defines the spatial reference and coordinate system.
5. **.sbn** : Spatial index file used to improve display and query performance.
6. **.sbx** : Companion spatial index file used together with .sbn
7. **.cpk** : Character encoding file that defines how text fields are stored (important for non-English languages).
8. **.shp.xml** : Metadata file that stores descriptive information about the dataset.

Shapefile VS Feature Class

Feature Class	Shapefile	Aspect
Stored inside a geodatabase	Multiple files on disk	Storage
1 TB	2 GB	Storage Limits
Supported	Not supported	Domains & Subtypes Attribute Rules & Topology
65,534	1024	Maximum number of fields
128	10	Maximum field name length
Supported (Enterprise GDB)	Not supported	Multi-user Editing
Professional GIS projects	Data exchange & simple projects	Use Cases

Import / Export Features

Data Conversion

Import / Export Features

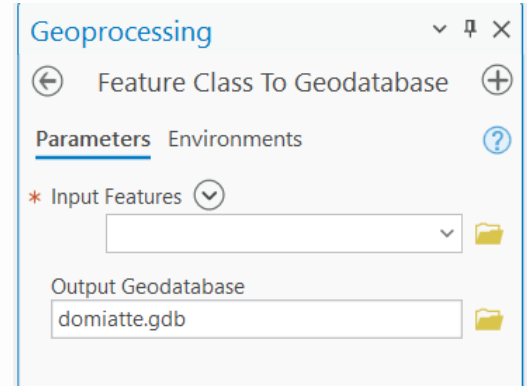
Not all GIS data is created from scratch. In many projects, spatial data comes from external sources and needs to be imported into a geodatabase , or exported to be shared with other systems or users.

Import and export operations allow GIS data to move between different formats, software, and workflows while preserving its geographic meaning.

Data can be imported into or exported from a **Geodatabase** or a **Feature Dataset**

using multiple methods :

- Drag and Drop
- Using the Geodatabase (Context Menu)
- Using a Feature Dataset
- Using Geoprocessing Tools



Data Conversion

In ArcGIS Pro, data conversion is performed using geoprocessing tools, primarily from the Conversion Toolbox, to transform GIS data between different formats (such as CAD to Geodatabase, KML to Layer, or CSV to Points) or to change data structures (such as converting features to points or lines)



Lab exercise

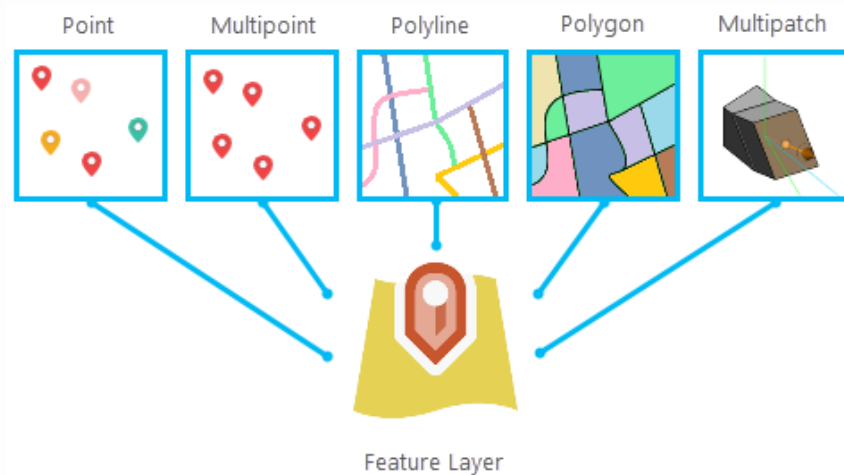
Using what you have learned in previous lessons, perform the following tasks

- Create a new **mosaic dataset** inside a geodatabase, **import** the DEM rasters,
- Create **Buildings** and **Roads feature classes**
- Create a **feature dataset** and import the previously created feature classes into it.
- Create a Tree **shapefile** and import it into the **geodatabase**
- Export the **Buildings** feature class as a **shapefile**

Feature Layer

Feature Layer (.lyrx)

A feature layer is how ArcGIS represents geographic features from a data source, such as a feature class or a shapefile. It provides a way to visualize, query, symbolize, and edit geographic data without changing the underlying data structure



Layer Properties

General : Controls general layer settings such as name, visibility, and scale-based display.

Metadata : Stores descriptive information about the layer, including source, author, and purpose.

Source : Displays the data source details, including storage location, fields, and spatial reference.

Elevation : Controls how the layer is positioned vertically in 3D scenes using elevation settings.

Selection : Defines how features are selected and highlighted within the layer.

Display : Controls feature display behavior on the map, such as reference scale symbol behavior.

Cache : Improves drawing performance by storing temporary display information.

Definition Query : Filters the layer to display only features that meet specified conditions without modifying the data.

Time : Enables time-based visualization using date or time fields.

Range : Allows dynamic filtering and visualization based on numeric attribute ranges.

Indexes : Manages spatial and attribute indexes to improve query and display performance.

Joins : Temporarily combines attributes from another table based on a common field.

Relates : Creates a logical relationship between tables without merging their attributes.

Page Query : Controls which features are displayed on each page in a map series or layout.

Attribute Table

Attribute Table - Definition

Attribute table is like a spreadsheet that stores non-spatial data describing geographic features.

Each **row** represents a feature (**record**), and each **column** represents an attribute (**field**)

The attribute table is linked to the map, allowing users to view, edit, query, and symbolize geographic data, with selections synchronized between the table and the map

[Row/Record]

[Column/Field]

[Cell]

OBJECTID	SHAPE	Name of Building	Size of Building	Operational Hours	Access
1	Polygon	Gym	1768.992398	8:30am-5:00pm	Emergency
2	Polygon	Gym	1043.515552	8:30am-5:00pm	Emergency
3	Polygon	J	3834.247963	8:30am-5:00pm	Emergency
4	Polygon	OA	14550.305355	8:30am-5:00pm	Emergency
5	Polygon	G	11507.921772	8:30am-5:00pm	Emergency
6	Polygon	S	2690.197171	8:30am-5:00pm	Emergency
7	Polygon	L	11246.731494	8:30am-5:00pm	Emergency
8	Polygon	E	10625.150848	8:30am-5:00pm	Emergency
9	Polygon	F	4913.179712	8:30am-5:00pm	Emergency
10	Polygon	O	25929.942586	8:30am-5:00pm	Emergency
11	Polygon		2740.696448	8:30am-5:00pm	Emergency

Search 0 of 27 selected

Attribute Table – Fields Management

Managing fields in an attribute table is a critical step in geodatabase design.

Field management defines how feature properties are stored, validated, and used in editing and analysis.

Well-designed fields improve data quality, reduce errors, and support advanced GIS workflows.

Fields are managed using the Fields view, which allows you to create, delete, and modify field properties for layers and tables

Visible	Read Only	Field Name	Alias	Data Type	Allow NULL	Highlight	Number Format	Domain	Default	Precision	Scale	Length
☒	☒	OBJECTID	OBJECTID	Object ID	☐	☐	Numeric			10	0	
☒	☐	BUILDINGID	Building Identifier	Text	☒	☐				0	0	50
☒	☐	FACILITYKEY	Facility Site Identifier	Text	☒	☐				0	0	50
1	☒	SHORTNAME	Name of Building	Text	☒	☐				0	0	25
☒	☐	BLDGAREA	Size of Building	Double	☒	☐	Numeric			38	8	
☒	☐	ACCESSTYPE	Access Type	Text	☒	☐				0	0	50
☒	☐	CONDITION	Condition	Text	☒	☐		Condition	2	0	0	50
☒	☐	OWNEDBY	Owned By	Short	☒	☐	Numeric	AssetOwner	Private	5	0	
☒	☐	MAINTBY	Managed By	Short	☒	☐	Numeric	AssetManager		5	0	
☒	☒	GlobalID	GlobalID	Global ID	☐	☐				0	0	
☒	☐	Shape	Shape	Geometry	☒	☐				0	0	
4	☒	OFFICEHOURS	Operational Hours	Time Only	☒	☐				0	0	
☒	☐	OPENDATE	Opening Date	Date Only	☒	☐				0	0	
☒	☒	Shape.STArea()	Shape.STArea()	Double	☒	☐	Numeric			0	0	
☒	☒	Shape.STLength()	Shape.STLength()	Double	☒	☐	Numeric			0	0	
5	☒	BLDGTYPE	Type of Campus Building	Short	☒	☐				5		

Click here to add a new field.

Field Type

Fields are the building blocks of feature classes and tables, and each field is defined by a data type. The data type determines what kind of values a field can store and how the data can be analyzed. Choosing the correct data type improves analysis results, system performance, and data integrity.

Text : Any set of alphanumeric characters of a specified length

Date and Time : Can be one of the following date or time data types:

- *Date* : Date and time values
- *Date only* : Date values only with no time values
- *Time only* : Time values only with no date values
- *Timestamp offset* : Date, time, and offset from UTC value

BLOB : Binary large objects are used to store documents, photographs, or other large data.

Global identifiers : unique value to identify a feature or table row across geodatabases

- *GUID* : user-defined field , can be created and managed manually **{6FFF-8B6-D01}**
- *Global ID* : A system-managed field automatically created and maintained by ArcGIS

Raster : data type can store the raster data in or alongside the geodatabase.

[Read More](#)

Numerical Field Type

Data type	Storable range	Size (bits)	Application
Short integer	-32,768 to 32,767	16	Numeric values without fractional values within a specific range coded values
Long integer	-2,147,483,648 to 2,147,483,647	32	Numeric values without fractional values within a specific range
Big integer	-9,007,199,254,740,991 to 9,007,199,254,740,991	64	Numeric values without fractional values within a specific range
Float	6-7 digits (total)	32	Numeric values with fractional values within a specific range
Double	16 digit (total)	64	Numeric values with fractional values within a specific range

Challenge



You want to create a field to store **building names** in a Buildings layer.
What is the most suitable data type?

Challenge



You want to create a field to store the **number of floors** for each building in a Buildings layer.

What is the most suitable data type?

Challenge



You want to create a field to store **tree heights (in meters)** in a Trees layer.

What is the most suitable data type?

Challenge



You want to create a field to store **coordinates (X or Y)** in a layer

What is the most suitable data type?

Challenge



You want to create a field to store **population count** for each district.

What is the most suitable data type?

Challenge



You want to create a field to store **only the inspection date** of a building (no time required).

What is the most suitable data type?

Challenge



You want to create a field to store the **exact date and time** of a traffic accident.

What is the most suitable data type?

Challenge



You want to create a field to store **binary attachments** related to land parcels.

What is the most suitable data type

Challenge



You want to create a field to store a **raster image** linked directly to a feature.

What is the most suitable data type?

Calculate field

Calculate Field is a tool used to batch update values in a table. Instead of typing data record by record, you use expressions to fill many rows at once.

Main Benefits :

- **Speed** : Update thousands of rows in a few seconds.
- **Accuracy** : Use mathematical formulas to avoid human errors.
- **Flexibility** : It works on numbers, text, and dates.

Common Uses :

- **Basic Math** : Perform calculations such as addition, subtraction, and multiplication using field values.
- **Text Formatting** : Combine or modify text values from multiple fields.
- **Logical Classification** : Apply if-then logic to categorize or classify data

Code Block : allows writing custom functions to handle complex logic that cannot be easily written in a single expression.

Calculate Geometry

Calculates geometric properties directly from feature geometry and writes the results into attribute fields, enabling spatial measurements such as length, area, coordinates (X, Y, Z), and M values to be stored, analyzed, and reused as attribute data

Spatial & Non-Spatial Tables

Spatial vs. Non-Spatial Tables

1. Non-Spatial Tables (Standalone Tables)

Definition : Tables that contain descriptive information (attributes) but **no geographic** shapes

Visual : They look like a standard Excel sheet

Content : Contains rows (records) and columns (fields) with data like names, dates, or numbers

Example : A list of customers with their phone numbers and total purchases

2. Spatial Tables (Attribute Tables)

Definition : Tables that are linked to a **Feature Class** (points, lines, or polygons).

Visual : Each row in the table is connected to a shape on the map.

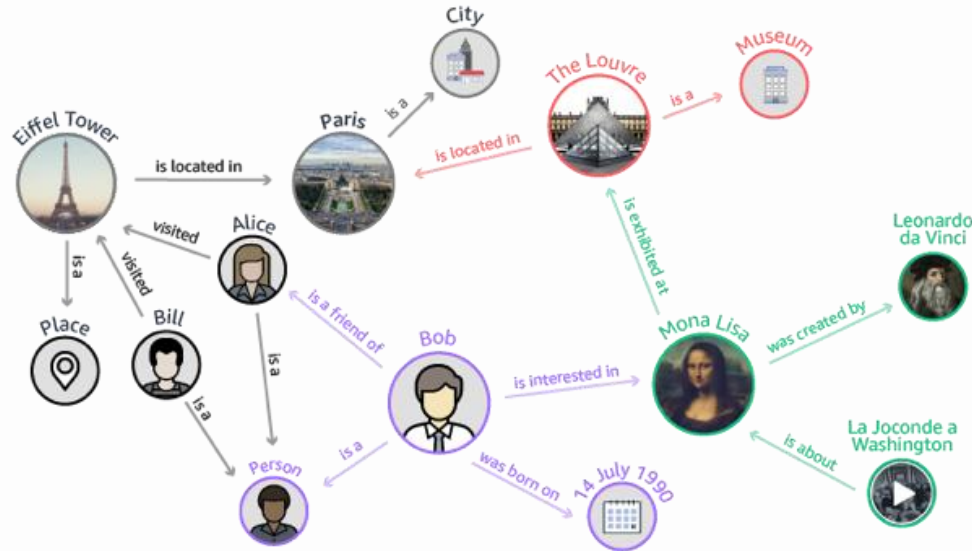
Key Feature : They contain a hidden or visible **"Shape" field** that stores the coordinates.

Example : A table of "Cities" where clicking a row highlights a point on the map.

Join & Relate

Data relationship options

Data relationships can be created in multiple ways, either **temporarily** by **joining** or **relating** tables within a map, or **permanently** by creating **relationship classes** in a **geodatabase**



Primary key & Foreign key

Primary keys and **Foreign keys** are fundamental database concepts used to establish and manage **relationships** between an origin table or feature class and a destination table. These keys ensure data integrity and enable efficient data retrieval

Primary Key (PK): A field in a table that uniquely identifies each record in that same table. It cannot contain duplicate values and cannot be null.

Foreign Key (FK): A field in one table (the destination or related table) that references the primary key in another table (the origin or parent table). It establishes a link between the records of the two tables. The foreign key field must have the same data type as the primary key it references



ParcelToPermit



Parcel (origin)

Parcel_ID	Zone	Block
123	...	...
456	...	...
789	...	...

**Origin
primary key**



Permit (destination)

Permit	Parcel_ID	FileDate
1	456	...
2	789	...
3	789	...
4	123	...

**Destination
foreign key**

Add Join

Add Join is a **temporary** table association that dynamically appends fields from a join table to an input table based on a common key field.

When a join is created, the joined information is stored at the layer level and exists only within the map or project where it was created.

The join can be saved with the project, but it is only visible to users who have access to that project

Join Requirements :

- Both tables must contain a common key field
- The key field names may differ, but the data type must be the same

Supported Cardinality

- One-to-One (1:1)
- Many-to-One (M:1)

Add join merges tables temporarily for viewing and analysis, but does not modify the original data

National Parks

OBJECTID	Unit Code	ParentOrgID
1	YELL	128
2	GRTE	128

+

OrganizationsTable

OBJECTID	OrgParentID	OrgType	OrgName
1	126	Federal Agency	Bureau of Land Management
2	127	Federal Agency	Fish and Wildlife Service
3	128	Federal Agency	National Park Service
4	129	Federal Agency	Bureau of Reclamation

Join
(M-1)

National Parks

OBJECTID	Unit Code	ParentOrgID	OrgParentID	OrgType	OrgName
1	YELL	128	128	Federal Agency	National Park Service
2	GRTE	128	128	Federal Agency	National Park Service

Add Relate

Add Relate establishes a logical association between two tables using a common key field, **without** physically **joining** the **tables**.

In a relate, tables remain **separate** however, when a record or feature is selected in one table, the related records in the other table are automatically selected.

Key Characteristics of Relate

- Tables are **not physically joined**
- Data remains stored in separate tables
- Related records are accessed through **selection**

Supported Cardinality

- One-to-One (1:1)
- One-to-Many (1:M)
- Many-to-Many (M:N)



CampgroundFacilities to Campsites

Campground Facilities X

OBJECTID	Shape	FacilityID	Facility Name
19	Point	259303	Grant Village Campground
20	Point	259305	Madison Campground
21	Point	259307	Pebble Creek Campground
22	Point	259308	Tower Fall Campground
23	Point	259309	Lewis Lake Campground
24	Point	259310	Slough Creek Campground
25	Point	259315	Mammoth Campground

Origin

Destination

Campsites X

OBJECTID	Shape	Campsite ID	ParentFacilityID	Campsite Name	Campsite Type
25	Point	10068712	259307	Pebble Creek site #25	STANDARD NONELECTRIC
26	Point	10068713	259307	Pebble Creek site #26	STANDARD NONELECTRIC
27	Point	10068714	259307	Pebble Creek site #27	MANAGEMENT
28	Point	10068715	259310	Slough Creek site #1	TENT ONLY NONELECTRIC
29	Point	10068716	259310	Slough Creek site #2	TENT ONLY NONELECTRIC
30	Point	10068717	259310	Slough Creek site #3	TENT ONLY NONELECTRIC
31	Point	10068718	259310	Slough Creek site #4	TENT ONLY NONELECTRIC
32	Point	10068719	259310	Slough Creek site #5	TENT ONLY NONELECTRIC
33	Point	10068720	259310	Slough Creek site #6	TENT ONLY NONELECTRIC
34	Point	10077219	259315	Mammoth site #1	MANAGEMENT
35	Point	10077220	259315	Mammoth site #2	STANDARD NONELECTRIC
36	Point	10077221	259315	Mammoth site #3	STANDARD NONELECTRIC
37	Point	10077222	259315	Mammoth site #4	STANDARD NONELECTRIC

Primary Key

Foreign Key

Relationship Classes

Relationship Classes

Relationship Class is a **geodatabase dataset** that defines a **permanent relationship** between related records

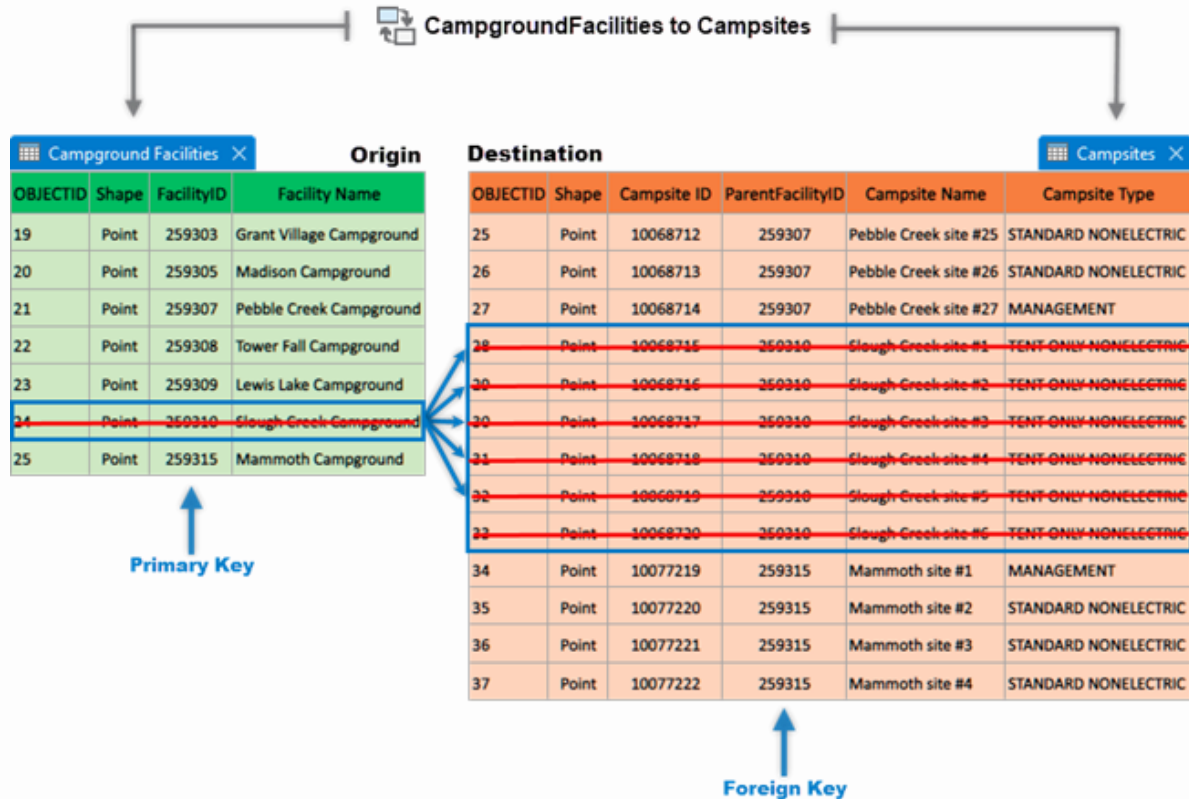
Key Characteristics

- Physically stored inside the geodatabase
- Defined once and reused across projects
- Enforced at the database level, not just the map or layer level

A relationship class is based on a primary key-foreign key relationship and supports the following cardinalities :

- One-to-One (1:1)
- One-to-Many (1:M)
- Many-to-Many (M:N)

Relationship classes can be configured to automatically update related records during editing



Lab exercise

Using what you have learned in previous lessons, perform the following tasks

- Using **Salzburg_Landuse** Feature class , Create **Definition Query** To display Only (**residential**) **LANDUSE**
- Add new Price **Range** to **States_Hotels** Feature class
- Calculate the **X and Y coordinates** for each hotel in the **States_Hotels** Feature class
- Add a new field named **Total_Income** to the **States_Hotels** Feature class and calculate the **total income** based on the number of rooms and room price
- Create a **temporary join** between the **States** Feature class and the **States_Data** table
- Create **permanent relationship** between **states** and **States_Hotels** Feature class

Outline | Subtype & Domain



01

Subtype

02

Domain

03

04

05

06

Subtype & Domain

Subtype

Subtypes are a way to divide features within a single feature class or table into logical categories based on a specific field.

They allow you to manage different types of features that share the same structure but have different behavior

Why Use Subtypes?

- Categorize features within the same feature class
- Assign default values to fields for each subtype
- Prevent data entry mistakes during editing
- Improve geodatabase performance by reducing the need for multiple feature classes

Requirement

- Subtypes must be based on a field with an **Integer data type**
- Each integer value represents a specific subtype

The screenshot shows the 'Manage Subtypes' dialog box. On the left, a table lists fields with their data types and domains. The 'BLDG_TYPE' field is highlighted. The dialog box has a 'Subtype Field' dropdown set to 'BLDG_TYPE' and a 'Default Subtype' dropdown set to 'Development'. Below these, a 'Subtypes' list shows four subtypes: 1 Development, 2 Utility, 3 Marketing, and 4 Security. The 'Subtype 1' row is selected. At the bottom, there is a 'Discover codes' button and 'OK' and 'Cancel' buttons.

Field Name	Data Type	Domain	Default Value
OBJECTID	Object ID		
Shape	Geometry		
Shape_Length	Double		
Shape_Area	Double		
BLDG_TYPE	Text		
BUILDING_ID	Long	1	
FACILITY_KEY	Text		
NAME_SHORT	Text		
BUILDING_AREA	Double		
ACCESS_TYPE	Text	Access_Type	Empl
CONDITION	Text	Condition	
OWNED_BY	Short	Asset_Owner	
MAINT_BY	Short	Asset_Manager	
LAST_UPDATE	Date		

Subtypes

Code	Name
1	Development
2	Utility
3	Marketing
4	Security

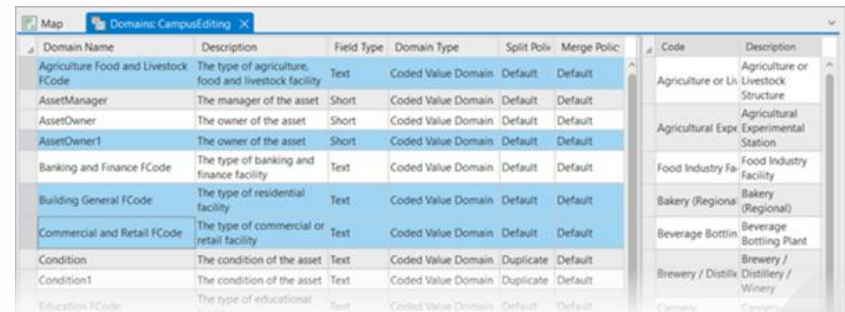
Domains

Domains define the valid values that can be entered into a field. Only **values** or **ranges** that are considered valid are accepted when a domain is applied.

Domains play a critical role in **maintaining data integrity**, especially when data is collected or edited by multiple users

There are two main types of domains that can be assigned to a field

1. **Coded Value Domain** : defines a fixed list of acceptable values for a field
Each value consists of
 - **code** (stored in the database)
 - **description** (displayed to the user)
2. **Range Domain** : defines a **minimum** and **maximum** value for **numeric** or **date** fields Any value outside this range is rejected.



Domain Name	Description	Field Type	Domain Type	Split Policy	Merge Policy	Code	Description
Agriculture Food and Livestock FCode	The type of agriculture, food and livestock facility	Text	Coded Value Domain	Default	Default	Agriculture or Livestock Structure	Agriculture or Livestock Structure
AssetManager	The manager of the asset	Short	Coded Value Domain	Default	Default	Agricultural Experimental Station	Agricultural Experimental Station
AssetOwner	The owner of the asset	Short	Coded Value Domain	Default	Default	Food Industry Facility	Food Industry Facility
AssetOwner1	The owner of the asset	Short	Coded Value Domain	Default	Default	Bakery (Regional)	Bakery (Regional)
Banking and Finance FCode	The type of banking and finance facility	Text	Coded Value Domain	Default	Default	Beverage Bottling Plant	Beverage Bottling Plant
Building General FCode	The type of residential facility	Text	Coded Value Domain	Default	Default	Brewery / Distillery / Winery	Brewery / Distillery / Winery
Commercial and Retail FCode	The type of commercial or retail facility	Text	Coded Value Domain	Default	Default	Cannery	Cannery
Condition	The condition of the asset	Text	Coded Value Domain	Duplicate	Default		
Condition1	The condition of the asset	Text	Coded Value Domain	Duplicate	Default		
Education FCode	The type of educational facility	Text	Coded Value Domain	Default	Default		

Lab exercise

Using what you have learned in previous lessons, perform the following tasks

- Use the **Boundary** layer to identify and inspect the study area
- Create new **File Geodatabase** to store the project data.
- Create a **Feature Dataset** with the appropriate spatial reference
- Create the required **feature classes** inside the feature dataset
- Design the **attribute fields** for each feature class.
- Define appropriate **Subtypes** to categorize features within the same feature class.
- Create and apply suitable **Domains** (coded value and/or range domains) to the fields.
- Assign **domains and default values per subtype** where applicable

Outline | Attribute Rules & Topology Rules



01

Attribute Rules

02

Constraint Rules

03

Calculation Rules

04

Validation Rules

05

Topology Rules

06

Georeferencing

Attribute Rules

Attribute Rules definition

Attribute Rules enforce data quality and integrity by automatically calculating values (**Calculation Rules**), restricting invalid edits (**Constraint Rules**), and identifying data quality issues (**Validation Rules**)

Rules are written using **Arcade expressions**, providing more advanced control than domains and subtypes.

Attribute Rules can be applied to **feature classes and tables**, and are managed through the **Catalog** or **Contents** pane under **Data Design** .

They help maintain data consistency during editing and support quality validation at a later stage

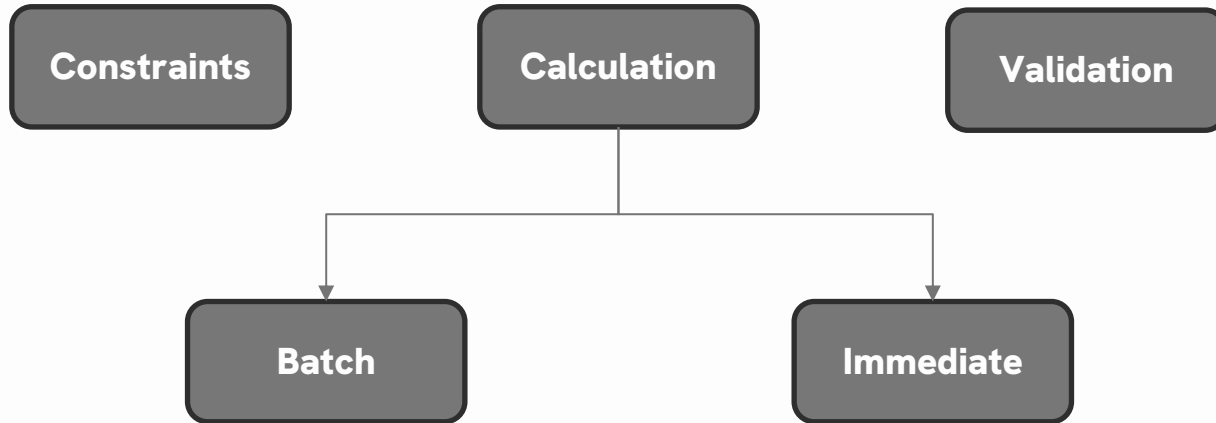
Why Attribute Rules Are Important ?

Attribute Rules act as the **bridge** between attribute structure and spatial integrity. They allow the geodatabase to :

- Automatically calculate attribute values
- Prevent invalid edits based on business logic
- Detect data quality issues without blocking editing
- Reduce manual calculations and human errors

Attribute Rules complement domains and subtypes, but **do not replace them**. Each mechanism works together to produce reliable, high-quality GIS data.

Attribute Rules Types



Constraint Rules



Attribute Rules enforce data quality by preventing invalid edits in real time, ensuring that features meet defined conditions such as value ranges, correct formatting, and spatial relationships.

When a rule is violated, a user-defined error message is triggered, preventing the edit from being saved.

When a user attempts to insert or update data, the rule automatically validates the input.

If the data satisfies the rule conditions, the edit is saved successfully. If the conditions are not met, the operation fails and a clear error message is presented to the user.

Calculation Rules



Calculation attribute rules are used to automatically populate or update attribute values in a feature or table.

They execute based on specific conditions and are written using Arcade expressions.

These rules help ensure

- Data consistency
- Reduced manual editing
- Accurate and standardized attribute values

Types of Calculation Attribute Rules :

- **Immediate** : Evaluated instantly **during edit** operations , Triggered by **Insert Update Delete**
- **Batch** : Evaluated at a user-defined time, **not during editing** , especially useful for large datasets

Evaluation Order : controls the sequence in which calculation rules are executed.

This is critical when One calculated field depends on another Changing the evaluation order may change the final calculated results

Validation Rules

Validation attribute rules help verify the integrity and correctness of data within a geodatabase by checking its compliance with defined rules and conditions.



These rules play a key role in

- 1. Error Detection**

Validation rules search for invalid, incorrect, or non-compliant values based on predefined criteria.

- 2. Data Review**

Validation rules do not prevent incorrect data from being entered , Instead, they validate the data after insertion or modification and flag errors for later review and correction

- 3. Improving Data Quality**

They assist users in ensuring that data conforms to established business rules, leading to higher overall data quality

New Rule

Calculation

Rule Name

A unique name that identifies the attribute rule

Description

Optional text explaining the purpose of the rule for documentation.

Subtype

☒ All

Specifies which subtypes the rule applies to , Can be applied to all subtypes or specific ones only

Field

☒ Editable

Field : The target field that will be calculated or updated by the rule.

Editable : Controls whether users can manually edit the field value . (**unchecked** = **system-maintained field**)

* Expression ☒

The calculation logic written in Arcade that defines how the field value is generated.

▼ * Triggers

Insert ☐ Update ☐ Delete ☐

Triggers : Defines when the rule is executed

> Triggering Fields

Specifies which field changes will trigger the rule during updates

▼ Execution

☐ Exclude from application evaluation

Exclude from application evaluation : Rule runs only in geodatabase evaluation, not during editing.

☐ Batch

Batch : Rule runs only when evaluated manually

☐ Disabled

Disabled : Temporarily turns the rule off

▼ Error

Error Number

A custom error code returned if the rule fails

Error Message

The message displayed to the user when the rule fails.

Tags

Keywords used to organize and search for rules.

Constraint Rules VS Validation Rules

Validation Rules	Constraint Rules
Do not prevent data entry, but identify errors in the data	Prevent data entry or modification immediately if a rule is violated
Used to check data quality after data has been entered or modified	Enforced instantly during the editing process
Require manual execution or scheduled validation	Executed automatically in real time
Allow edits but flag violations for later review	Stop invalid edits before they are saved

Lab exercise

Using Attribute Rules, determine the appropriate rule type to implement the following requirements

- Automatically calculate the **building height** based on floor field , and populate the **Height** field during **insert and update** operations
- Automatically calculate the Construction Year For New buildings during **insert** operations
- Prevent the creation of **Shade Trees** outside parcels classified as **Garden**

Topology Rules

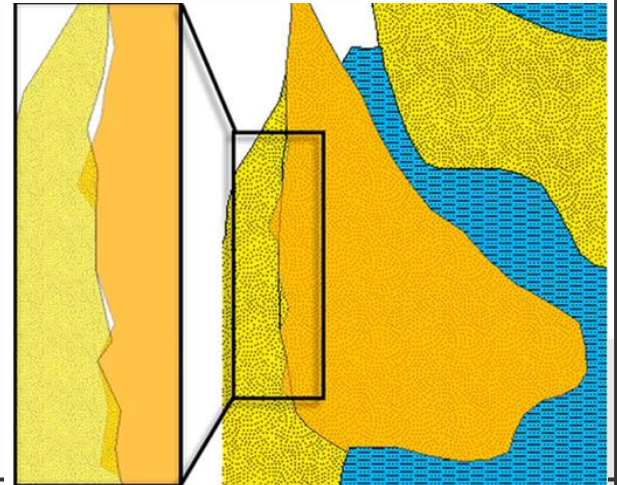
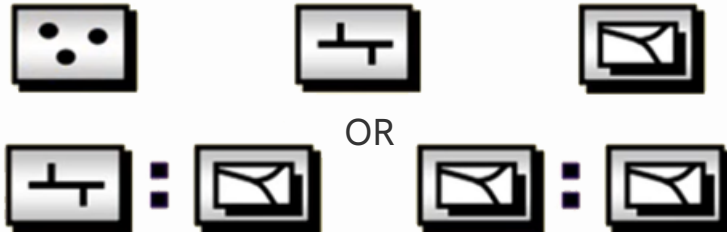
Topology Rules

Topology Rules enforce spatial integrity by controlling how features share geometry, connect, overlap, or interact with each other.

They help to prevent spatial errors such as overlaps, gaps, and disconnected features, and can automatically detect and correct digitizing mistakes.

Topology rules can be applied between :

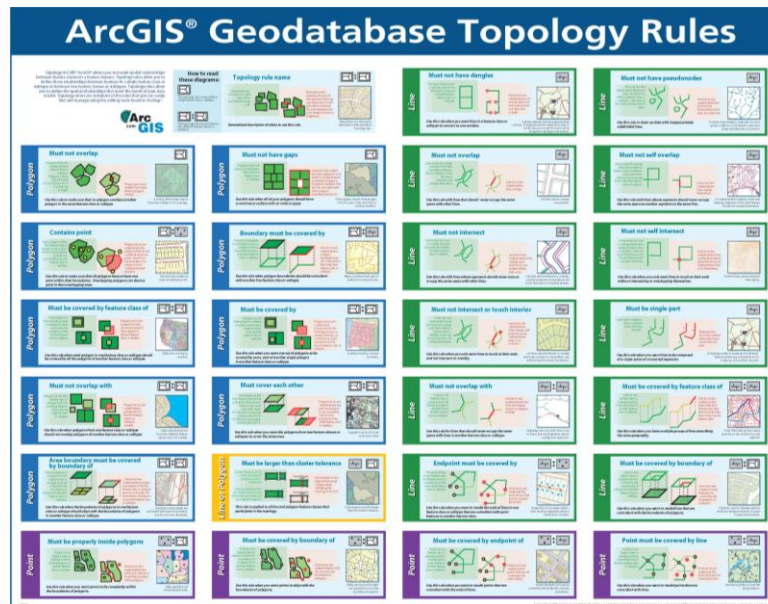
- Features within the **same feature class**
- Features from **different feature classes**
- Features belonging to **specific subtypes** of feature classes



Topology Rules

Examples of common topology rules applied to different geometry types

- **Must Not Have Gaps (Polygon)**
Ensures complete spatial coverage with no missing areas
- **Must Not Self-Intersect (Line)**
Prevents invalid geometry that causes analysis errors
- **Must Be Disjoint (Point)**
Ensures points are spatially independent

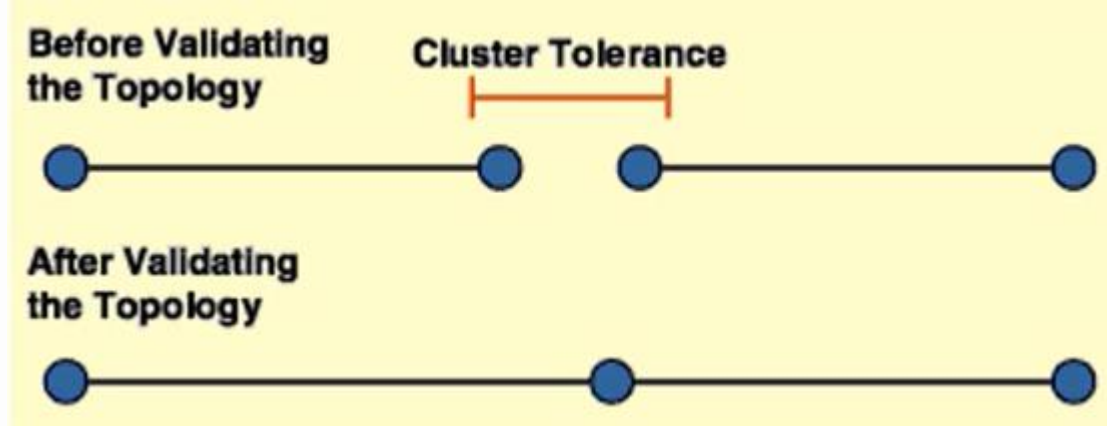


→ [Download](#)

Cluster Tolerance in Topology Rules

is the distance used by the geodatabase topology engine to determine when vertices from different features represent the same location.

Vertices that fall within this tolerance are **snapped together** and assigned a common coordinate during topology validation



Ranks in Topology Rules

Ranks control how features are adjusted during topology validation.

Lower rank values (for example, **Rank 1**) indicate **higher positional accuracy**, while **higher rank** values indicate **less accurate** data.

When features from different feature classes fall within the cluster tolerance, features with higher ranks (less accurate) snap to features with lower ranks (more accurate).

This ensures spatial integrity by prioritizing reliable data, such as surveyed features, over less accurate sources

Dirty Areas and Topology Validation

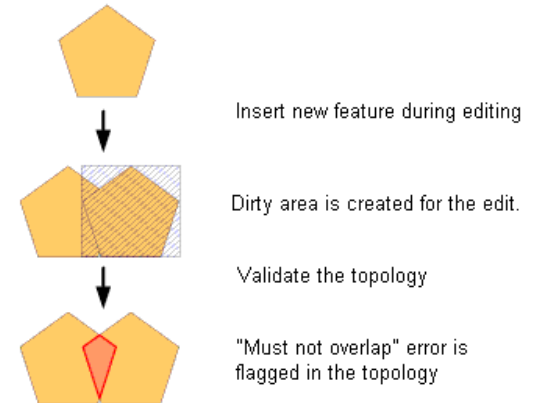
Topology Validation :

Topology validation is a **user-driven process** used to check whether features comply with defined topology rules or not . The Feature classes remain usable whether the topology is validated or not and validation can be performed after every edit or at the end of an editing session

Topology Dirty Areas :

locations where features have been **created, modified, or deleted** since the last topology validation , its allow ArcGIS to

- Track where changes occurred
- Validate only the affected areas instead of the entire dataset
- Improve performance and reduce processing time



Errors and Exceptions

Topology Errors :

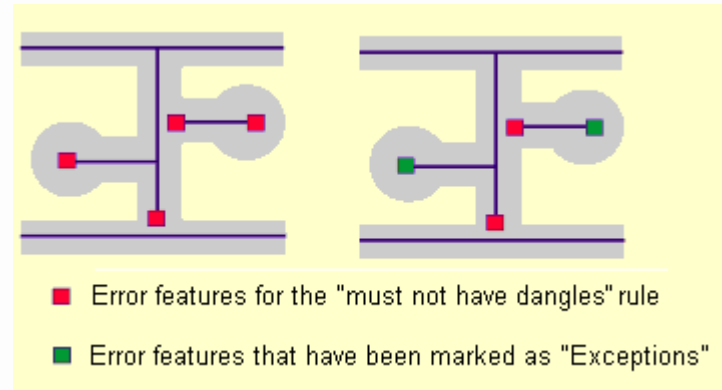
Topology rule violations are initially recorded as **errors** during topology validation.
Error features indicate where a feature does not comply with a defined topology rule

Topology Exceptions :

Some topology errors may be **acceptable in real-world scenarios**.

In such cases, errors can be marked as **exceptions**.

- Exceptions are ignored during future validations
- They remain documented in the topology
- They can be restored back to errors if needed



Data Quality Pyramid in Geodatabase

- **Field Type** : Defines the basic nature of data (Text, Integer, Double, Date).
Prevents storing incompatible values.
- **Domains** : Restrict allowed values or ranges for a field
Ensures standardized and valid attribute input.
- **Subtypes** : Categorize features within the same feature class.
Allow different default values and domain assignments.
- **Attribute Rules** : Automate calculations, enforce business logic, and validate data quality.
Provide advanced control beyond domains and subtypes.
- **Topology Rules** : Ensure spatial integrity between features.
Control spatial relationships such as overlap, gaps, connectivity, and containment

Lab exercise

Create a topology dataset and apply the appropriate topology rules to the following layers

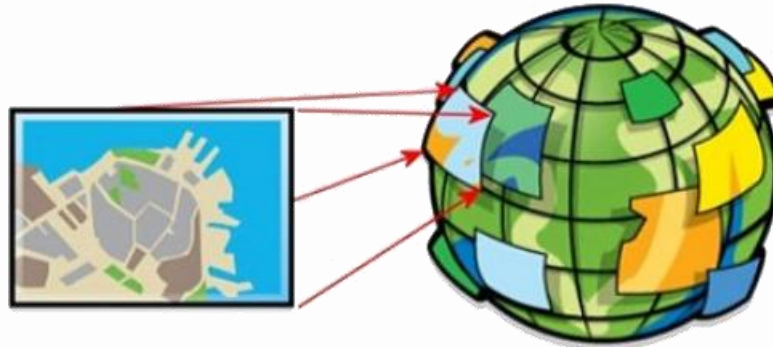
- Parcel
- Trees
- Buildings
- Roads

The applied rules should ensure valid geometry and correct spatial relationships within the editing area.

Georeferencing

Georeferencing

Georeferencing is the process of aligning an unreferenced image (raster) or CAD file to a known coordinate system by creating Ground Control Points (GCPs) that link locations on the image to real-world geographic coordinates

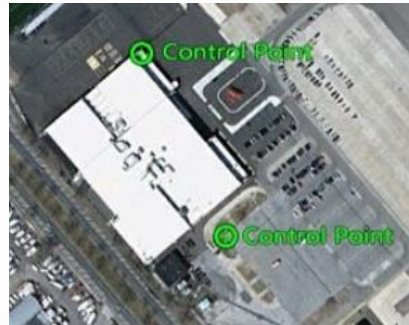


Control Points (GCPs)

Control Points, also known as **Ground Control Points (GCPs)**, are known reference locations used to link positions on an unreferenced image to their correct real-world geographic coordinates.

Each control point connects :

- A point on the image (pixel location)
- To a known location in a coordinate system (map location)



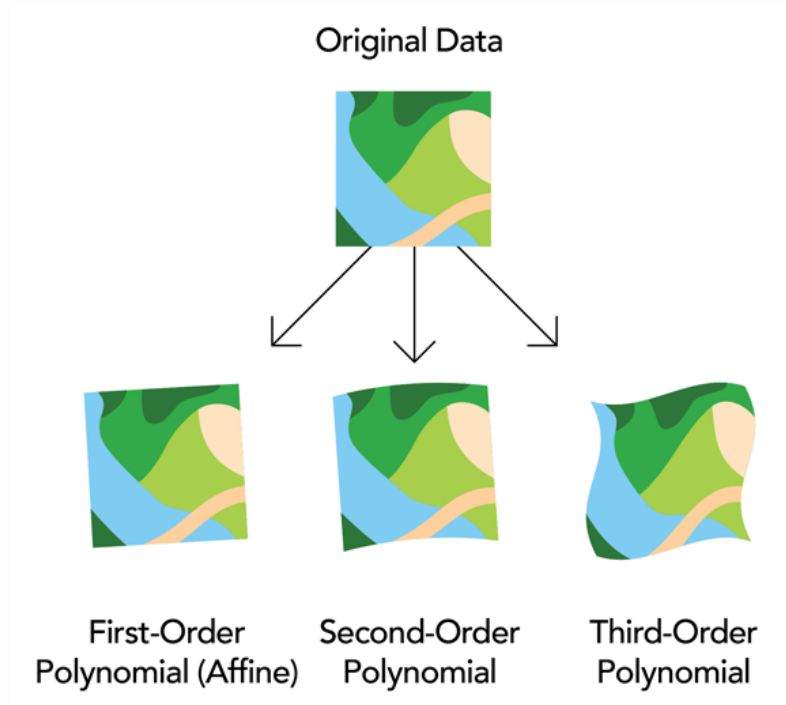
Transformations Types

Transformations define how an image or raster is mathematically adjusted to align with real-world coordinates using control points.

Each transformation type handles distortion differently and requires a specific number of control points

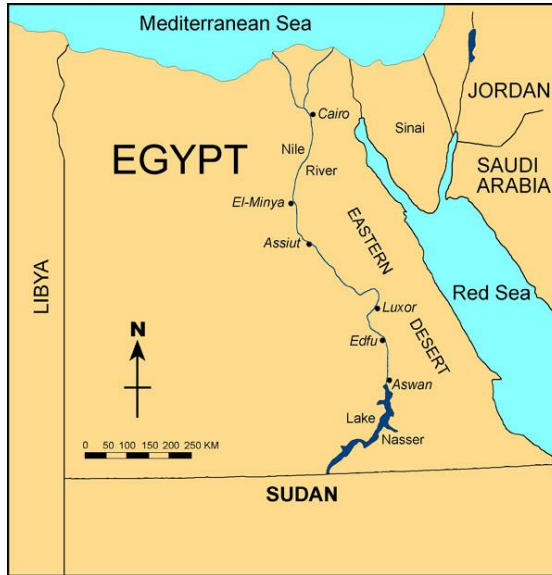
- **Shift (Zero-Order Polynomial)** : Required **1 control points** , Moves the image without rotating or scaling
- **Affine (1st-Order Polynomial)** : Required **3 control points** , Shifts, rotates, scales, and skews the image
- **Polynomial (2nd & 3rd Order)** : Required **6 control points for 2nd order & 10 for 3rd order** , Bends and warps the image to handle complex distortion
- **Adjust** : Required **3 control points** , Combines polynomial transformation with local adjustments
- **Spline** : Required **10 control points** , Exactly matches control points, **optimizing local accuracy**
- **Projective** : Required **4 control points** , Preserves perspective and projection geometry

Transformations Types



Lab exercise

Using the georeferencing techniques learned previously, perform a georeferencing process for the image shown below to align it with the correct real-world coordinate system



→ [Download](#)

Digitizing Workshop

Thanks!

Do you have any questions?

Kareem Ashraf

Information Technology Institute |
Geoinformatics Department

kaashraf@iti.gov.eg

